Lab 6: Value of Information

CEVE 421/521

2026-02-27

Draft Material — This content is under development and subject to change.

1 Overview

In Labs 4 and 5 you found optimal flood defenses for a city. Today we switch scales: a **homeowner in Norfolk, VA** must decide how high to elevate their house above the floodplain.

The key uncertainty is **sea-level rise (SLR)**, which depends on the global emissions pathway. Two futures are possible:

1. **Aggressive mitigation (RCP 2.6)** — the world cuts emissions sharply, SLR stays modest.
2. **Business as usual (RCP 8.5)** — emissions continue growing, SLR accelerates.

The homeowner doesn't know which pathway the world will follow, but must decide on an elevation height **now** (construction happens once). Today you'll compute the **Expected Value of Perfect Information (EVPI)** and the **Expected Value of Imperfect Information (EVII)** to determine how much it's worth to learn about future emissions before committing.

References:

- SimOptDecisions documentation — the optimization framework
- K. L. Ruckert, V. Srikrishnan, and K. Keller [1] — BRICK sea-level rise projections
- M. Zarekarizi, V. Srikrishnan, and K. Keller [2] — house elevation cost model

2 Before Lab

Complete these steps BEFORE coming to lab:

1. **Accept the GitHub Classroom assignment** (link on Canvas)
2. **Clone your repository:**

   ```bash
   git clone https://github.com/CEVE-421-521/lab-06-S26-yourusername.git
   cd lab-06-S26-yourusername
   ```

3. **Open the notebook** in VS Code or Quarto preview
4. **Run the first code cell** — it will automatically install any missing packages (this may take 10-15 minutes the first time)

! Important

Submission: You will render this notebook to PDF and submit the PDF to Canvas. See Section 8 for details.

i Note

Computation note: This lab uses deterministic trapezoidal integration (not Monte Carlo), so all cells should run in a few seconds.

3 Setup

3.1 Load Packages

```
# install SimOptDecisions
try
    using SimOptDecisions
catch
    import Pkg
    try
        Pkg.rm("SimOptDecisions")
    catch
    end
    Pkg.add(; url="https://github.com/dossgollin-lab/SimOptDecisions")
    Pkg.instantiate()
    using SimOptDecisions
end
```

```
using CairoMakie
using CSV
using DataFrames
using Distributions
using Interpolations
using NetCDF
using Random
using Statistics

include("house_elevation.jl")
```

3.2 Build the House

We use a HAZUS depth-damage function for a single-story house with no basement, typical of Norfolk's coastal housing stock. The house sits at 8 ft above the local tide gauge.

```
haz_fl_dept = CSV.read("data/haz_fl_dept.csv", DataFrame)

# Single-story, no basement, A-Zone (FIA DmgFnId 105)
```

```
ddf_row = filter(r -> r.DmgFnId == 105, haz_fl_dept)[1, :]

house = House(
    ddf_row;
    value_usd=250_000,      # house value
    area_ft2=1_500,         # floor area
    height_above_gauge_ft=8.0,  # first-floor elevation above tide gauge
)
```

3.3 Sea-Level Rise from BRICK

We load BRICK sea-level rise projections [1] for Sewells Point, Norfolk, VA. The BRICK ensemble contains thousands of SLR trajectories under each RCP scenario.

The two states of the world correspond to the **emissions pathway** the world follows:

1. **RCP 2.6** — aggressive mitigation (10th percentile of BRICK ensemble)
2. **RCP 8.5** — business as usual (90th percentile of BRICK ensemble)

We use tail percentiles (not ensemble means) to sharpen the contrast: RCP 2.6 with a low climate response, vs RCP 8.5 with a high climate response.

```
const M_TO_FT = 3.28084

slr_data = let
    slr_nc = joinpath(@__DIR__, "data", "sea_level_projections.nc")
    brick_years = Int.(ncread(slr_nc, "time"))
    brick_slr = Float64.(ncread(slr_nc, "brick_slr"))  # (time, ensemble, rcp)
    rcps = ["rcp26", "rcp45", "rcp60", "rcp85"]
    (years=brick_years, slr=brick_slr, rcps=rcps)
end

function brick_percentile(data, rcp, p)
    idx = findfirst(==(rcp), data.rcps)
    ensemble = data.slr[:, :, idx]
    return [quantile(ensemble[t, :], p) for t in axes(ensemble, 1)]
end

slr_low_m = brick_percentile(slr_data, "rcp26", 0.10)   # optimistic: low emissions +
low sensitivity
slr_high_m = brick_percentile(slr_data, "rcp85", 0.90)   # pessimistic: high
emissions + high sensitivity
```

```
# Prior: 50/50 chance of high vs low emissions
p_high = 0.5
p_low = 1 - p_high

# Fixed parameters
surge_dist = GeneralizedExtremeValue(3.0, 1.5, 0.1)  # annual max storm surge (ft)
discount_rate = 0.04
```

```julia
years = collect(2026:2100)  # 75-year planning horizon

# Map BRICK years to our planning horizon and convert to feet
brick_year_start = findfirst(==(2026), slr_data.years)
brick_year_end = findfirst(==(2100), slr_data.years)
slr_low_ft = slr_low_m[brick_year_start:brick_year_end] .* M_TO_FT
slr_high_ft = slr_high_m[brick_year_start:brick_year_end] .* M_TO_FT

# Build configs and scenarios
config = HouseConfig(house, years, surge_dist)
scenario_low = SLRScenario(slr_low_ft, discount_rate)
scenario_high = SLRScenario(slr_high_ft, discount_rate)
```

```julia
let
    yr = slr_data.years
    fig = Figure(; size=(700, 350))
    ax = Axis(
        fig[1, 1];
        xlabel="Year",
        ylabel="Sea Level Rise (ft)",
        title="Sea-Level Rise: RCP 2.6 (P10) vs RCP 8.5 (P90)",
    )
    lines!(ax, yr, slr_low_m .* M_TO_FT; color=:blue, linewidth=2, label="RCP 2.6 P10
(optimistic)")
    lines!(ax, yr, slr_high_m .* M_TO_FT; color=:red, linewidth=2, label="RCP 8.5 P90
(pessimistic)")
    vlines!(ax, [2026, 2100]; color=:gray, linestyle=:dot, alpha=0.5)
    axislegend(ax; position=:lt)
    fig
end
```

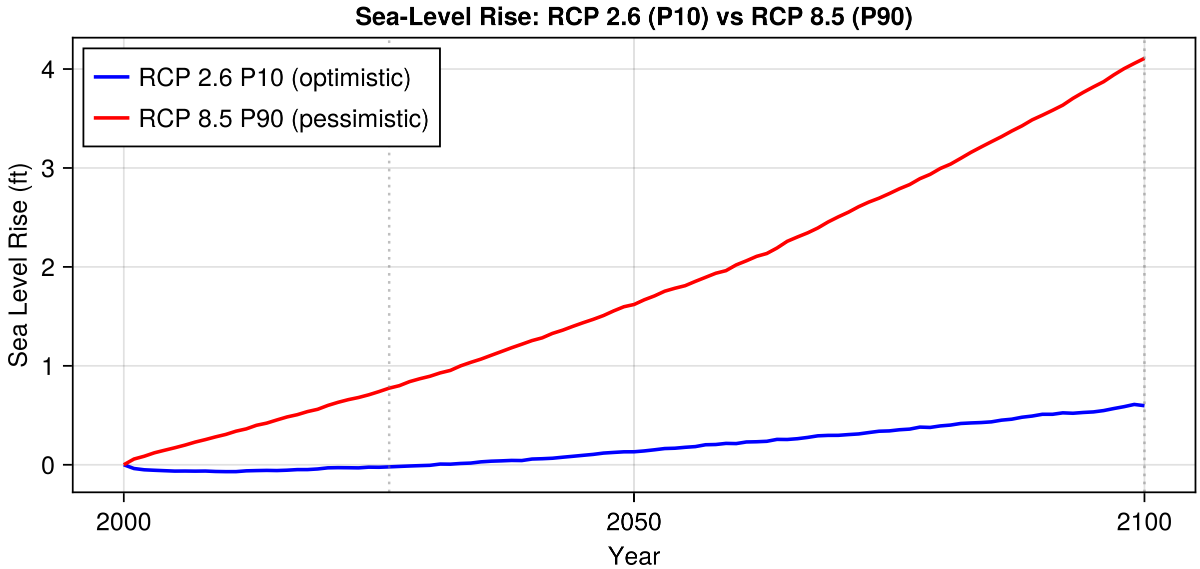

Figure 1: BRICK SLR projections. Blue: RCP 2.6, 10th percentile (optimistic). Red: RCP 8.5, 90th percentile (pessimistic). Dashed lines mark the planning horizon.

4 Build the Payoff Table

We sweep a grid of elevation heights and compute total cost (construction + NPV of expected flood damages) under each SLR future.

```
height_grid = 0.0:0.5:14.0

costs_low = [compute_total_cost(config, scenario_low, h) for h in height_grid]
costs_high = [compute_total_cost(config, scenario_high, h) for h in height_grid]
costs_prior = p_high .* costs_high .+ p_low .* costs_low
```

```
let
    fig = Figure(; size=(700, 450))
    ax = Axis(
        fig[1, 1];
        xlabel="Elevation Height (ft)",
        ylabel="Total Cost (USD)",
        title="Payoff Table: Elevation Decision Under SLR Uncertainty",
    )

    hg = collect(height_grid)
    lines!(ax, hg, costs_low; color=:blue, linewidth=2, label="RCP 2.6 (mitigation)")
    lines!(ax, hg, costs_high; color=:red, linewidth=2, label="RCP 8.5 (business as
usual)")
    lines!(ax, hg, costs_prior; color=:black, linewidth=2, linestyle=:dash,
label="Prior (50/50)")

    # Mark optima
```

```
    idx_low = argmin(costs_low)
    idx_high = argmin(costs_high)
    idx_prior = argmin(costs_prior)

    scatter!(ax, [hg[idx_low]], [costs_low[idx_low]]; color=:blue, markersize=15,
marker=:star5)
    scatter!(ax, [hg[idx_high]], [costs_high[idx_high]]; color=:red, markersize=15,
marker=:star5)
    scatter!(ax, [hg[idx_prior]], [costs_prior[idx_prior]]; color=:black,
markersize=15, marker=:star5)

    axislegend(ax; position=:rt)
    fig
end
```

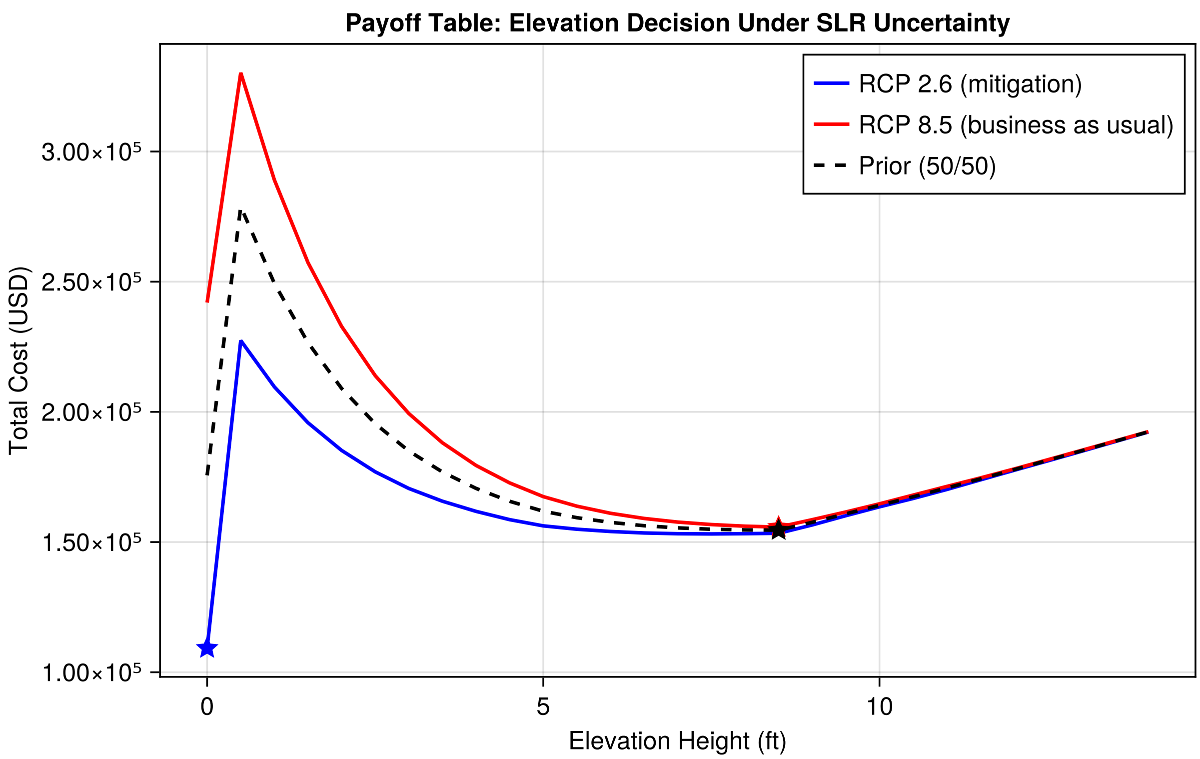

Figure 2: Total cost (construction + NPV of expected damages) vs. elevation height under each emissions pathway. Blue: RCP 2.6. Red: RCP 8.5. Dashed black: prior-weighted average (50/50). Stars mark the optimum for each curve.

Your Response — Question 1

Look at Figure 2.

1. *How do the optimal elevation heights differ between the two SLR futures? Why does this make intuitive sense?*
2. *Why are the cost curves U-shaped? What happens at very low vs. very high elevation?*

5 Computing EVPI

Use the payoff table to compute the Expected Value of Perfect Information. Since we're minimizing cost (not maximizing utility), the formula uses min instead of max:

$$\text{EVPI} = \underbrace{\min_a \sum_s p(s) \cdot C(a,s)}_{\text{cost without info}} - \underbrace{\sum_s p(s) \cdot \min_a C(a,s)}_{\text{cost with perfect info}}$$

where a is the elevation height, s is the SLR state, $p(s)$ is the prior probability, and $C(a,s)$ is total cost.

```
# Your code here
```

Table 1: EVPI for the elevation decision under SLR uncertainty

```
# Your code here
```

Your Response — Question 2

Look at Table 1.

1. *Interpret the EVPI in dollars. How much would this homeowner pay for a perfect forecast of the future emissions pathway?*
2. *What fraction of the house value ($250,000) does the EVPI represent?*
3. *Is this EVPI large enough to justify funding climate policy research? What other homeowners might benefit from the same information?*

6 Computing EVII

Suppose an international **Climate Policy Progress Report** rates the likelihood of meeting Paris Agreement goals as either "on track" or "off track." The report isn't perfectly accurate: it tends to say "on track" when emissions are following RCP 2.6, and "off track" when following RCP 8.5, but it sometimes gets it wrong.

6.1 Signal Accuracy

```
p_offtrack_given_85 = 0.85    # P(report = "off track" | RCP 8.5)
p_ontrack_given_26 = 0.90     # P(report = "on track" | RCP 2.6)
```

6.2 Exercise: Bayes' Rule

Use Bayes' rule to compute the posterior probability of RCP 8.5 given each report signal.

```
# Your code here
```

6.3 Exercise: EVII

Re-weight the payoff table using the posterior probabilities and compute EVII. No new simulations needed — just re-weight the same cost arrays with updated beliefs.

$$\text{EVII} = \underbrace{\min_a \sum_s p(s) \cdot C(a,s)}_{\text{cost without info}} - \underbrace{\sum_z p(z) \min_a \sum_s p(s \mid z) \cdot C(a,s)}_{\text{cost with signal}}$$

where z is the report signal and $p(s \mid z)$ is the posterior from Bayes' rule.

```
# Your code here
```

Table 2: Summary of information value: no information, policy report, and perfect information

```
# Your code here
```

```
# Your code here
```

Figure 3

Your Response — Question 3

What fraction of the EVPI does the policy report capture? Why is EVII < EVPI? What would make the report more valuable?

Your Response — Question 4

The homeowner is considering two research investments:

1. *Fund a climate policy analysis that would perfectly predict the future emissions pathway.*
2. *Fund a tide gauge network that would reduce storm surge uncertainty.*

Based on your EVPI result, what is the maximum value of investment (1)? Why might investment (2) have a different (possibly larger or smaller) ceiling?

Extensions (Optional)

If you finish early, try one of these:

- **Change the prior:** What happens to EVPI if RCP 8.5 is 80% likely instead of 50/50? At what prior does EVPI become negligible?

- **Change signal accuracy:** What if the policy report is only 60% accurate? 99% accurate? Plot EVII vs. accuracy.
- **More RCP scenarios:** Add RCP 4.5 and RCP 6.0 as additional states. How does EVPI change with more emissions pathways?

7 Summary

Key takeaways from this lab:

1. **EVPI** is the gap between "optimize then average" and "average then optimize" — the maximum value of any information source about a given uncertainty.
2. **EVII** extends EVPI to imperfect signals via Bayes' rule — re-weight the same payoff table with posterior beliefs, no new simulations needed.
3. **Information is only valuable when it changes the optimal decision.** If the optimal action is the same regardless of the state, EVPI is zero.
4. The EVPI for one uncertainty bounds the value of any research into that uncertainty — use it to prioritize research investments.

8 Submission

1. Write your answers in the response boxes above.
2. **Render to PDF:**
   - In VS Code: Open the command palette (`Cmd+Shift+P` / `Ctrl+Shift+P`) → "Quarto: Render Document" → select Typst PDF
   - Or from the terminal: `quarto render index.qmd --to typst`
3. **Submit the PDF** to the Lab 6 assignment on Canvas.
4. **Push your code** to GitHub (for backup):

```
git add -A && git commit -m "Lab 6 complete" && git push
```

Checklist

Before submitting:

- ☐ Packages load without error
- ☐ SLR trajectories plotted (Figure 1)
- ☐ Payoff table visualized (Figure 2)
- ☐ Question 1 answered (interpret cost curves)
- ☐ EVPI computed (Table 1)
- ☐ Question 2 answered (interpret EVPI)
- ☐ Bayes' rule posteriors computed
- ☐ EVII computed and summary table generated (Table 2)
- ☐ EVII visualization (Figure 3)
- ☐ Questions 3 and 4 answered

- ☐ Notebook renders to PDF without errors

Bibliography

[1] K. L. Ruckert, V. Srikrishnan, and K. Keller, "Characterizing the Deep Uncertainties Surrounding Coastal Flood Hazard Projections: A Case Study for Norfolk, VA," *Scientific Reports*, vol. 9, no. 1, pp. 1–12, Aug. 2019, doi: 10.1038/s41598-019-47587-6.

[2] M. Zarekarizi, V. Srikrishnan, and K. Keller, "Neglecting Uncertainties Biases House-Elevation Decisions to Manage Riverine Flood Risks," *Nature Communications*, vol. 11, no. 1, p. 5361, Oct. 2020, doi: 10.1038/s41467-020-19188-9.